

SLAM EX4

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[Github](#)

Additional to EX3:

We evaluated multiple detector/descriptor configurations and PnP variants. Although AKAZE achieved the best matching performance and the highest inlier ratio (0.968), its computational cost was considerably higher. Since the visual odometry pipeline was required to process thousands of frames, ORB was chosen as the final detector due to its favorable balance between tracking quality and runtime efficiency. This choice enabled full-sequence processing in approximately 205 seconds while still maintaining robust pose estimation.

```
ORB 700:    218 inliers / 261 correspondences = 0.835
ORB 1000:   298 inliers / 362 correspondences = 0.823
ORB 1500:   421 inliers / 554 correspondences = 0.760
AKAZE:      456 inliers / 471 correspondences = 0.968
```

4.1:

We implemented a custom TrackingDB class that maintains the tracking information using several hash maps (Python dictionaries). The database stores a mapping from each TrackId to the list of frames in which the track appears (track_to_frames), and a mapping from each FrameId to the set of tracks visible in that frame (frame_to_tracks). Feature observations are stored in an Observation object containing the stereo coordinates (x_l, x_r, y) (x_l, x_r, y) (x_l, x_r, y), and are indexed by the pair (FrameId, TrackId) through the observations dictionary. To efficiently extend tracks across consecutive frames, the database also maintains a feature_to_track dictionary that maps a feature location (FrameId, FeatureIdx) to its corresponding TrackId. During database construction, temporal matches obtained in Exercise 3 are used to either extend an existing track or create a new one, allowing tracks to grow naturally as features are successfully tracked across multiple stereo pairs. This design provides constant-time access to tracks, frames, and observations while remaining simple and easy to serialize for later stages such as Bundle Adjustment and Loop Closure.

4.2:

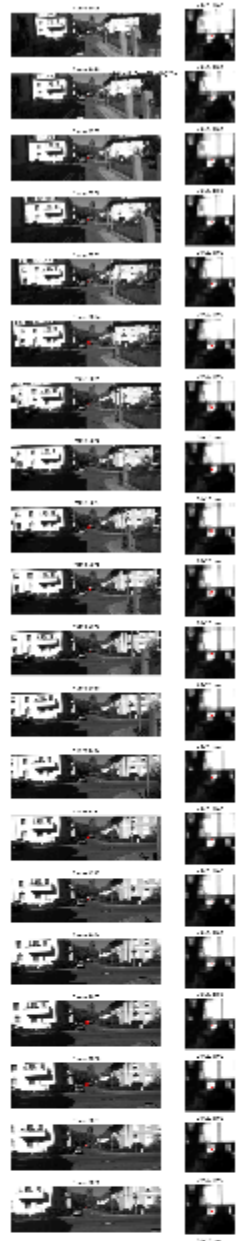
Tracking Statistics (50 Frames)

- Total number of tracks: **259723**
- Number of frames: **3300**
- Mean track length: **3.33 frames**
- Maximum track length: **66 frames**
- Minimum track length: **2 frames**
- Mean number of frame links: **261.95 tracks per frame**

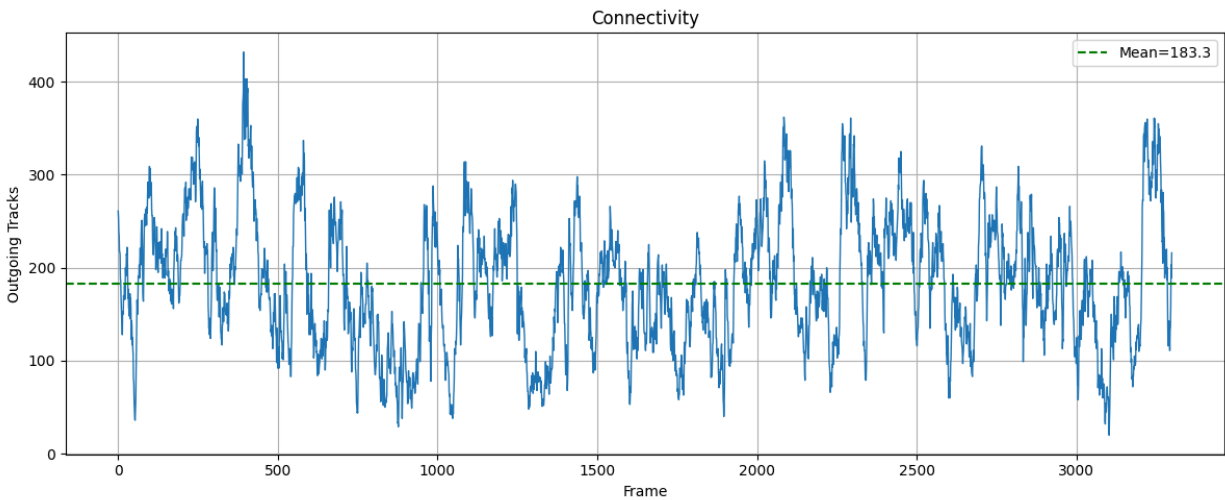
After removing trivial tracks of length 1, the database contained **259,723 tracks**. The average track length was **3.33 frames**, while the longest track successfully persisted for **66 consecutive frames**. The minimum track length was 2 frames, as expected from the track construction process. On average, each frame contained **261.95 tracked landmarks**, indicating that a substantial number of visual features were successfully maintained throughout the sequence despite viewpoint changes, occlusions, and tracking failures.

4.3:

A track of length **66 frames** (track id: 157279) was selected and visualized across all of its observations. For each frame, the feature location was marked on the left image and a 20×20 image patch centered on the feature was extracted. The selected landmark remained visible throughout frames 2069–2134 and exhibited a highly consistent local appearance across the entire track. The stability of both the image patches and the tracked location demonstrates the ability of the tracking pipeline to maintain long-term correspondences over extended portions of the sequence. Such long tracks are particularly valuable for later stages such as Bundle Adjustment and Loop Closure, where persistent landmarks provide strong geometric constraints.

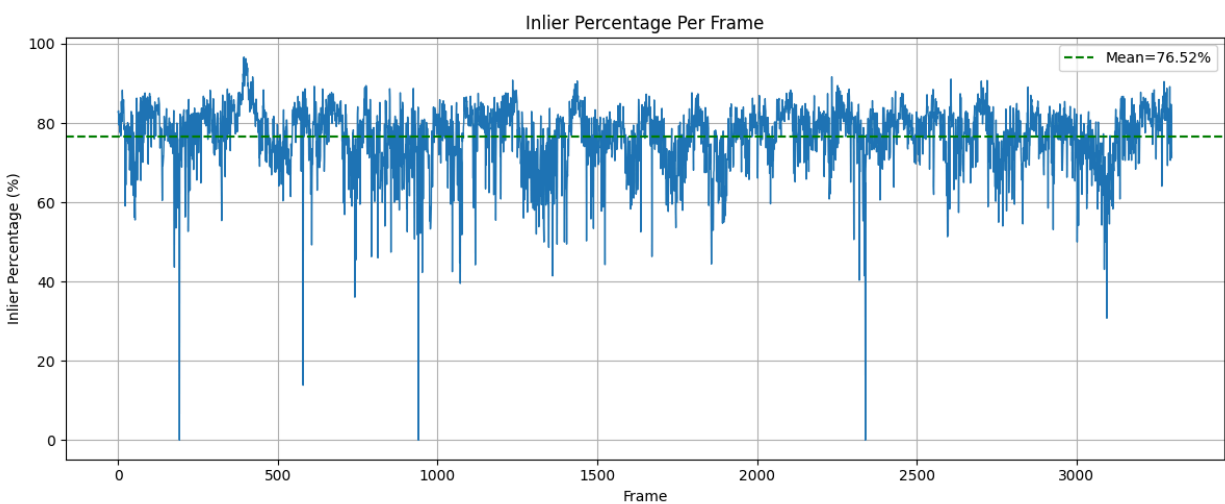


4.4:



The connectivity graph shows, for each frame, the number of tracks that continue to the following frame. The average connectivity over the entire sequence was 183.31 outgoing tracks per frame. Although significant fluctuations are visible throughout the trajectory, the graph indicates that a large number of landmarks are consistently tracked between consecutive frames. Variations in connectivity are expected and are mainly caused by scene changes, occlusions, viewpoint variations, and differences in image texture richness across the sequence.

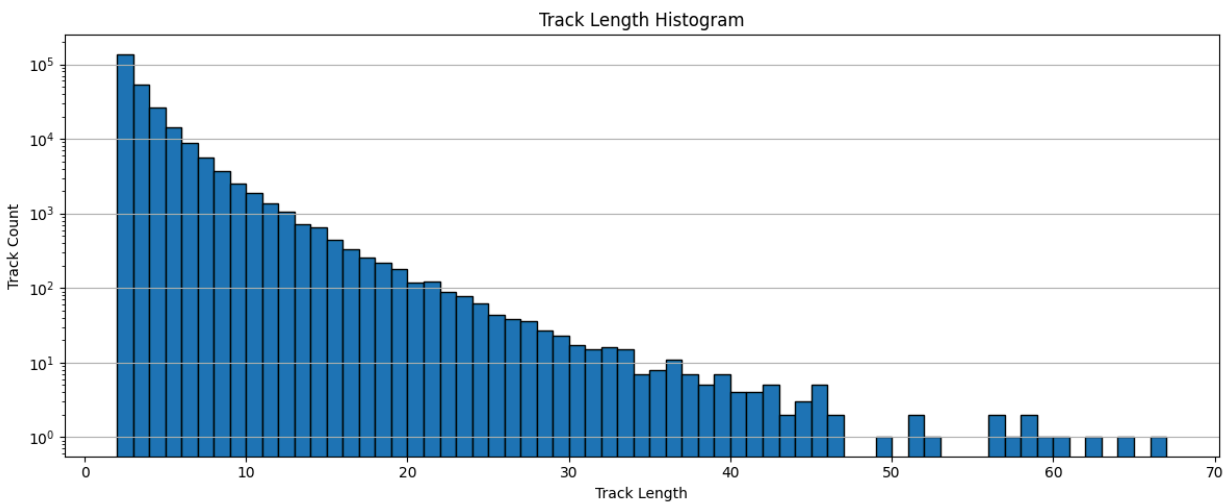
4.5:



For each pair of consecutive frames, we computed the percentage of PnP-RANSAC inliers out of all candidate correspondences. The resulting graph shows the temporal

evolution of tracking quality throughout the sequence. The average inlier percentage was 76.52%, indicating that approximately three quarters of the tracked correspondences were geometrically consistent with the estimated camera motion. Most frames exhibit inlier ratios between 70% and 85%, demonstrating robust feature matching and pose estimation. Occasional drops in the inlier ratio correspond to challenging regions in the sequence, such as viewpoint changes, occlusions, or reduced image texture.

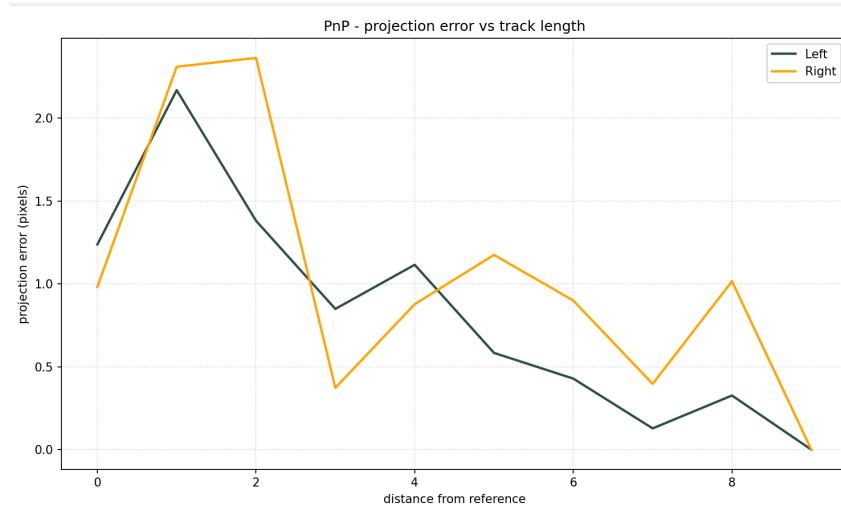
4.6:



The histogram presents the distribution of track lengths across the entire sequence. Most tracks are relatively short, while progressively fewer tracks survive for longer periods. This behavior is expected in visual tracking systems, since landmarks may leave the field of view, become occluded, or fail the matching process as the camera moves through the environment. The distribution exhibits a long-tailed shape, with a small number of highly persistent tracks surviving for dozens of consecutive frames. The longest track in the database remained active for 66 frames, demonstrating the ability of the tracking pipeline to maintain long-term feature correspondences when stable visual landmarks are present.

4.7:

1)



2) If the triangulation were performed using the first frame of the track instead of the last frame, the reprojection error graph would display a complete inversion of its trend (a mirror image across the X-axis). The error at the first frame would drop to zero, as it now serves as the reference frame to which the 3D point is mathematically optimized, and the error would gradually accumulate and increase as the vehicle moves forward in time toward the later frames.

This geometric behavior is driven by three main factors:

- a)** Optimization to the Reference Frame: The triangulation algorithm calculates the 3D coordinates (X, Y, Z) to maximize consistency with the pixel observations (x_l , x_r , y) of the reference frame. Consequently, the reconstructed point is mathematically "fitted" to that specific frame, resulting in a near-zero reprojection error when projected back onto it.
- b)** Camera Pose Drift: The ground-truth camera matrices accumulate small measurement errors and drift over time along the trajectory. When a 3D point is locked to the coordinate system of the first frame, the growing geometric discrepancy between the estimated and actual camera poses in subsequent frames translates into an increasing projection error on the 2D pixel sensor.
- c)** Sensitivity to Perspective and Distance Changes: As the vehicle advances, the viewing angles (ray baseline) and physical distance to the landmark change significantly. Small sub-pixel measurement noises introduced during the feature matching phase in the reference frame are mathematically magnified by sharp perspective shifts in distant frames, causing the projected rays to land further away from the actual tracked pixel locations.

4.8: (The mathematical sections of this document were prepared in LyX for better formatting and then transferred here.)

1)

Let $x \in R^n$ be a multivariate normal random vector such that $x \sim N(\mu_x, \Sigma_x)$, and let $y = Ax + b$ be a linear transformation where $A \in R^{n \times n}$ is a nonsingular matrix and $b \in R^n$ is a constant vector.

1. Proving the Mean $\mu_y = A\mu_x + b$

By applying the linearity property of the expectation operator $E[\cdot]$, we can derive the expected value of y :

$$\mu_y = E[y] = E[Ax + b]$$

Using the linearity of expectation, we can distribute the operator over the summation and pull out the constant matrix A and the constant vector b :

$$\mu_y = E[Ax] + E[b] = AE[x] + b$$

Since the mean of x is given as $E[x] = \mu_x$, substituting this into the equation yields:

$$\mu_y = A\mu_x + b$$

2. Proving the Covariance $\Sigma_y = A\Sigma_x A^T$

By definition, the covariance matrix of the random vector y is given by:

$$\Sigma_y = E[(y - \mu_y)(y - \mu_y)^T]$$

First, let us find an expression for the deviation vector $(y - \mu_y)$. Substituting $y = Ax + b$ and $\mu_y = A\mu_x + b$:

$$y - \mu_y = (Ax + b) - (A\mu_x + b) = Ax - A\mu_x = A(x - \mu_x)$$

Now, we substitute this deviation vector back into the covariance definition:

$$\Sigma_y = E[(A(x - \mu_x))(A(x - \mu_x))^T]$$

Applying the matrix transpose property $(CD)^T = D^T C^T$ to the second term inside the expectation:

$$\Sigma_y = E[A(x - \mu_x)(x - \mu_x)^T A^T]$$

Using the linearity of the expectation operator once again, we can pull out the deterministic constant matrices A (from the left) and A^T (from the right) outside of the expectation:

$$\Sigma_y = A \underbrace{E[(x - \mu_x)(x - \mu_x)^T]}_{\Sigma_x} A^T$$

Since the term in the middle is the definition of the covariance matrix of x (Σ_x), we obtain:

$$\Sigma_y = A\Sigma_x A^T$$

2)

To prove that $y \sim N(\mu_y, \Sigma_y)$, we utilize the multivariate random vector transformation formula. For a bijective transformation $y = g(x) = Ax + b$, the probability density function (pdf) of y is given by:

$$f_y(y) = f_x(g^{-1}(y)) \cdot |J_{g^{-1}}|$$

where $|J_{g^{-1}}|$ represents the absolute determinant of the Jacobian matrix of the inverse transformation $g^{-1}(y)$.

1. Finding the Inverse Transformation $g^{-1}(y)$

We begin with the direct transformation equation and isolate x step by step to find the inverse function:

$$y = Ax + b$$

First, we subtract the constant vector b from both sides of the equation to isolate the term containing x :

$$y - b = Ax$$

Since A is given as a nonsingular (invertible) matrix, its inverse A^{-1} exists. To isolate x completely, we multiply both sides of the equation from the left by A^{-1} :

$$A^{-1}(y - b) = A^{-1}(Ax)$$

Using the associative property of matrix multiplication, the right side of the equation can be rewritten as:

$$A^{-1}(y - b) = (A^{-1}A)x$$

Since multiplying a matrix by its inverse results in the identity matrix ($A^{-1}A = I$), and multiplying any vector by the identity matrix leaves it unchanged ($Ix = x$), we obtain the final expression for the inverse transformation:

$$x = g^{-1}(y) = A^{-1}(y - b)$$

2. Finding the Jacobian Determinant $|J_{g^{-1}}|$

The Jacobian matrix of the linear inverse function $g^{-1}(y) = A^{-1}y - A^{-1}b$ with respect to y is simply the matrix of coefficients:

$$J_{g^{-1}} = \frac{\partial g^{-1}(y)}{\partial y} = A^{-1}$$

Taking the absolute determinant of the Jacobian matrix yields:

$$|J_{g^{-1}}| = |A^{-1}| = \frac{1}{|A|}$$

3. Evaluating the Probability Density Function $f_x(g^{-1}(y))$

The known probability density function of the multivariate normal vector $x \sim N(\mu_x, \Sigma_x)$ is:

$$f_x(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_x|}} \exp \left(-\frac{1}{2} (x - \mu_x)^T \Sigma_x^{-1} (x - \mu_x) \right)$$

Substituting $x = A^{-1}(y - b)$ into $f_x(x)$, let us first simplify the term inside the exponent:

$$x - \mu_x = A^{-1}(y - b) - \mu_x = A^{-1}(y - b) - A^{-1}A\mu_x = A^{-1}(y - (A\mu_x + b))$$

Recalling from Part 1 that $\mu_y = A\mu_x + b$, we can substitute it into the expression:

$$x - \mu_x = A^{-1}(y - \mu_y)$$

Now, substituting this back into the full exponent expression yields:

$$(x - \mu_x)^T \Sigma_x^{-1} (x - \mu_x) = (A^{-1}(y - \mu_y))^T \Sigma_x^{-1} (A^{-1}(y - \mu_y))$$

Applying the transpose rule $(CD)^T = D^T C^T$:

$$= (y - \mu_y)^T (A^{-1})^T \Sigma_x^{-1} A^{-1} (y - \mu_y)$$

Using basic matrix properties, we know that $(A^{-1})^T = (A^T)^{-1}$. Thus:

$$= (y - \mu_y)^T (A^T)^{-1} \Sigma_x^{-1} A^{-1} (y - \mu_y)$$

By factoring the inverse out using the matrix reverse-order rule for inverses $((CDH)^{-1} = H^{-1}D^{-1}C^{-1})$:

$$= (y - \mu_y)^T (A \Sigma_x A^T)^{-1} (y - \mu_y)$$

Notice that the term inside the brackets is exactly the covariance matrix of y derived in Part 1 ($\Sigma_y = A \Sigma_x A^T$). Therefore, the exponent simplifies cleanly to:

$$= (y - \mu_y)^T \Sigma_y^{-1} (y - \mu_y)$$

4. Compiling the Final PDF $f_y(y)$

Now, we substitute our derived components back into the random vector transformation formula:

$$f_y(y) = f_x(g^{-1}(y)) \cdot |J_{g^{-1}}| = \left[\frac{1}{\sqrt{(2\pi)^n |\Sigma_x|}} \exp \left(-\frac{1}{2} (y - \mu_y)^T \Sigma_y^{-1} (y - \mu_y) \right) \right] \cdot \frac{1}{|A|}$$

To combine the terms, we move the scalar $\frac{1}{|A|}$ inside the square root denominator as $|A|^2$:

$$f_y(y) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_x| \cdot |A|^2}} \exp \left(-\frac{1}{2} (y - \mu_y)^T \Sigma_y^{-1} (y - \mu_y) \right)$$

Using determinant properties, specifically that $|A|^2 = |A| \cdot |A| = |A| \cdot |A^T|$, we get:

$$|\Sigma_x| \cdot |A|^2 = |A| \cdot |\Sigma_x| \cdot |A^T| = |A \Sigma_x A^T| = |\Sigma_y|$$

Substituting $|\Sigma_y|$ into the denominator results in:

$$f_y(y) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_y|}} \exp \left(-\frac{1}{2} (y - \mu_y)^T \Sigma_y^{-1} (y - \mu_y) \right)$$

This final expression is exactly the probability density function of a multivariate normal distribution with mean μ_y and covariance matrix Σ_y .

Hence, it is formally proven that $y \sim N(\mu_y, \Sigma_y)$.